KAIZEN MATHS

Lesson starter

Advanced Factorisation: Grouping, Difference of Squares and Cubics

Difference of Two Squares

Name _____

S1

$25 - m^2$

S2

$5x^2 - 180$

S3

$9 - (m + n)^2$

S4

$16x^4 - 81$

Show your working

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Exit ticket

Advanced Factorisation: Grouping, Difference of Squares and Cubics

Difference of Two Squares

Name _____

E1

$36 - p^2$

E2

$fm^2 - f$

E3

$(3x + 1)^2 - (6x - 1)^2$

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Teacher answers & working

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S1

$25 - m^2$

Answer:

$(5 - m)(5 + m)$

Working:

$A^2 - B^2 = (A - B)(A + B)$

$A = 5, B = m$

$A - B = 5 - m$

$A + B = 5 + m$

$25 - m^2 = (5 - m)(5 + m)$

S2

$5x^2 - 180$

Answer:

$5(x - 6)(x + 6)$

Working:

$5x^2 - 180 = 5(x^2 - 36)$

$= 5(x^2 - 6^2)$

Inside the bracket: A = x, B = 6

$A - B = x - 6$

$A + B = x + 6$

$= 5(x - 6)(x + 6)$

S3

$9 - (m + n)^2$

Answer:

$(3 - m - n)(3 + m + n)$

Working:

$9 - (m + n)^2 = 3^2 - (m + n)^2$

$A = 3, B = (m + n)$

$A - B = 3 - (m + n) = 3 - m - n$

$A + B = 3 + (m + n) = 3 + m + n$

$= (3 - m - n)(3 + m + n)$

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S4

$16x^4 - 81$

Answer:

$(2x - 3)(2x + 3)(4x^2 + 9)$

Working:

$16x^4 - 81 = (4x^2)^2 - 9^2$

First use: $A = 4x^2$, $B = 9$

$A - B = 4x^2 - 9$

$A + B = 4x^2 + 9$

$= (4x^2 - 9)(4x^2 + 9)$

Now factor $4x^2 - 9$: $A = 2x$, $B = 3$

$A - B = 2x - 3$

$A + B = 2x + 3$

$= (2x - 3)(2x + 3)(4x^2 + 9)$

E1

$36 - p^2$

Answer:

$(6 - p)(6 + p)$

Working:

$A^2 - B^2 = (A - B)(A + B)$

$A = 6, B = p$

$A - B = 6 - p$

$A + B = 6 + p$

$36 - p^2 = (6 - p)(6 + p)$

Teacher answers & working

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E2

$fm^2 - f$

Answer:

$f(m - 1)(m + 1)$

Working:

$fm^2 - f = f(m^2 - 1)$

$= f(m^2 - 1^2)$

Inside the bracket: A = m, B = 1

$A - B = m - 1$

$A + B = m + 1$

$= f(m - 1)(m + 1)$

E3

$(3x + 1)^2 - (6x - 1)^2$

Answer:

$9x(2 - 3x)$

Working:

$A = 3x + 1, B = 6x - 1$

$A - B = (3x + 1) - (6x - 1) = 2 - 3x$

$A + B = (3x + 1) + (6x - 1) = 9x$

$= 9x(2 - 3x)$